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MIL-DTL-32729/1
10 January 2023

DETAIL SPECIFICATION SHEET

CONVERTER, AIRCRAFT, 28 VOLTS DIRECT CURRENT (VDC), UNIVERSAL SERIAL BUS CHARGING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-32729.

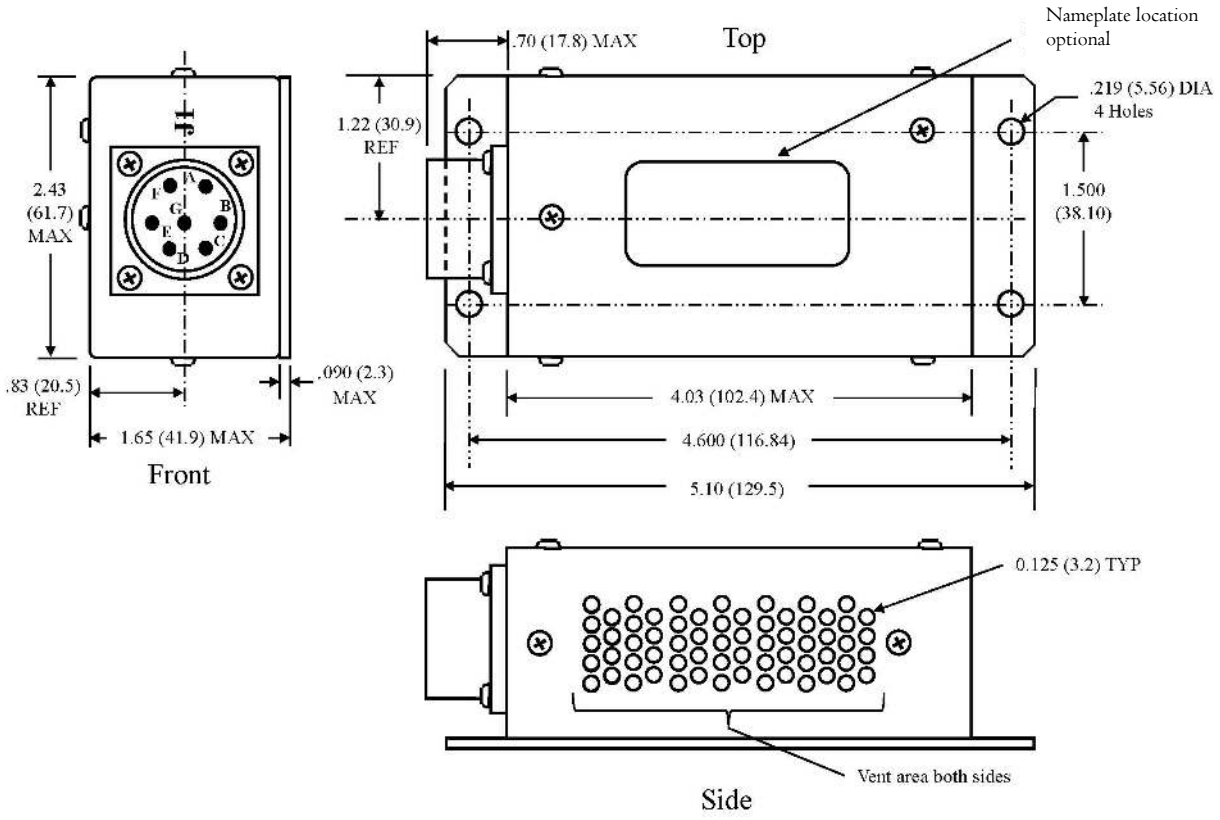


FIGURE 1. USB charging converter M32729/1-1.

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FIGURE NOTES

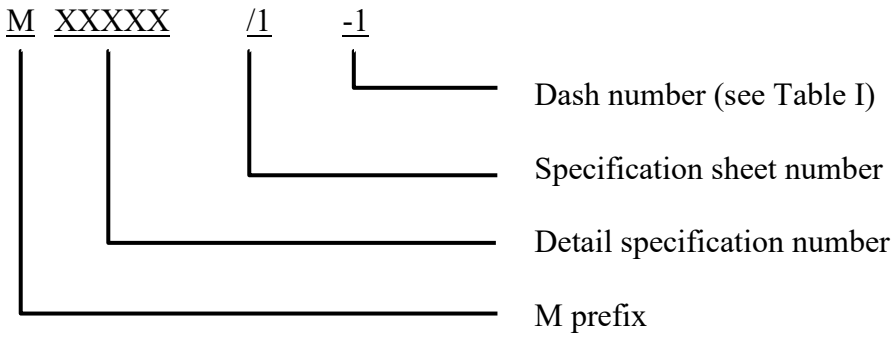
1. Unless otherwise specified, dimensions are in inches: fractional tolerance ±1/16; decimal tolerances ±0.1 for one decimal (0.0), ± 0.03 for two decimals, ± 0.010 for three decimals; angular tolerances ± 1/2°. Metric equivalents are shown in parentheses, are for reference only, and are based on 1 inch = 25.4 mm. Figure 1 dimensions exclude hardware.

REQUIREMENTS

1. Converter type. The converter type shall be Type DC.

2. Temperature/altitude rating. The converter shall provide the electrical performance over a temperature altitude range of -55 °C to +71 °C (-67 °F to 160 °F) in accordance with MIL-DTL-32729, temperature-altitude up to 55,000 ft. (16.8 km) altitude and as modified herein.

3. Part or identifying number (PIN) example:



4. Converter marking. The converter nameplate shall be located on the top of the converter, as shown on Figure 1.

5. Converter rating:

TABLE I. Converter description.

Part number	Power rating	Output	Max weight (pounds)	Efficiency
M32729/1-1	75 Watt	5 VDC 15 Amps	0.80 (0.36 KG)	78% minimum (at full load)

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6. External electric connections.

TABLE II. Connector contact assignments.

Connector	Contact	Description
J1 (Input/output) Pin contacts	A	Chassis ground
	B	Spare
	C	+5 VDC 1/
	D	+5 VDC 1/
	E	+5 VDC Return
	F	+28 VDC Input
	G	28 VDC Return

1/ Pins C and D may be externally electrically tied together for maximum current rating or separately connected to supply power to multiple USB charging ports.

7. Connectors. Connector J1 shall be a MS3102A16S-1P connector as specified in SAE AS31021 and oriented on Figure 1. The input and output connections and contact assignments shall be marked as shown in table II and on Figure 1.

8. Converter mounting. The converter shall include four mounting holes as shown on Figure 1. The converter, when mounted by the mounting screws, shall meet the vibration and shock requirements of this detail specification sheet.

9. Cooling. The converter may include venting holes (see Figure 1). The converter shall be convection cooled.

10. Color. The converter color shall be black in accordance with MIL-DTL-32729.

11. Size and weight. The size and dimensions of the converter shall be as shown on Figure 1. Converter weight shall be as required in table I. Figure 1 specifies the converter maximum dimensions.

12. Current returns. The 28 VDC and 5 VDC current returns shall be connected to their own current return wires. The current return wiring shall be separate from any wires used to bond internal components to the chassis of the converter.

13. Type AC converter input power. Not required.

14. Overload and short circuit capacity. Over an input voltage range of 22 to 29 VDC, the converter shall be capable of operating without damage to any passive load impedance.

a. Overload. Over the Type DC normal input voltage range, the converter shall maintain the minimum output voltage while delivering 150 percent rated output current for 15 seconds, without damage to the converter.

b. Short circuit current. The maximum short circuit current shall be 20 amperes maximum.

15. Overvoltage protection. The maximum overvoltage shall be 125 percent from no load to full load.

16. Efficiency. See Table I.

17. Electromagnetic compatibility. The converter shall meet the requirements of MIL-STD-461 Navy Rotary Aircraft equipment.

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18. Electrical bonding. Electrical bonding of the converter shall be through burnished areas around the four mounting holes of the converter (see Figure 1). Electrical bonding of the converter to the aircraft structure shall not exceed a maximum electrical DC resistance of 2.5 milliohms per MIL-STD-464. External metallic parts (connectors, housing, brackets and mounting flanges) shall be electrically bonded together with a maximum electrical resistance of 2.5 milliohms per interface per MIL-STD-464. Aluminum alloy forging surfaces shall be given a chemical film treatment in accordance with MIL-DTL-5541, Class 3 requirements to insure compliance to MIL-STD-464.

19. Safety. Converter safety design and operational safety requirements shall be in accordance with those applicable requirements specified in MIL-STD-882 system safety process.

20. Shock. The converter shall be subjected to functional shock (Procedure I). Crash hazard shock is not required. Shocks will be sawtooth pulse with a peak pulse of at least 20g's for 11 milliseconds.

21. Acoustical noise. Not required.

22. Insulation resistance. The converter shall have an insulation resistance of 10 meg-ohms minimum.

23. Exported transients. The converter shall not create transients (spikes) less than 50 microseconds in duration, which are greater than +50/-150 percent of the nominal 28 VDC input. Exported transient requirements apply to steady state operation and power on/off and off/on conditions.

24. Lightning induced transient susceptibility. The converter shall remain functionally operational during and after lightning induced electrical transients.

25. Spikes. The converter shall not be damaged by spikes of +/- 600V peak through a source impedance of 50+/-5 ohms on the 28 VDC converter input. The spikes shall have a duration of 50 microseconds maximum and may occur during all power conditions.

26. Water drip. The converter shall remain functionally operational during water drip tests and shall meet full performance output voltage requirements following the water drip test.

27. Toxicity and fire resistance. Not required.

VERIFICATION:

1. Airflow. Not applicable.

2. Converter input power. Only type DC input power is required. Type AC input power is not required.

3. Output load recovery. For 200 percent load and short circuit application and removal, tests shall be conducted at nominal input voltage, 22 VDC input, and 29 VDC input.

4. Voltage and frequency regulation at maximum temperature-altitude conditions. Maximum temperature-altitude limits shall be in accordance with Table V of MIL-DTL-32729 except the maximum temperature-altitude limit shall be -10 °C (14 °F) at 55,000 feet (16.8 km).

5. Voltage and frequency regulation at minimum temperature-altitude conditions. Minimum temperature-altitude limits shall be in accordance with Table V of MIL-DTL-32729 except the minimum temperature-altitude limit shall be -55 °C (-67 °F) at 55,000 feet (16.8 km).

6. External electromagnetic environment. The output voltage shall remain within the normal

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limits (see MIL-DTL-32729, Output Power) while the converter is exposed to MIL-STD-464 external electromagnetic environments. The external electromagnetic environment shall be a composite following MIL-STD-464 environments:

a. Maximum external electromagnetic environment for deck operations on Navy ships.

b. Maximum external electromagnetic environment for ship operations in the main beam of transmitters.

c. Maximum external electromagnetic environment for rotary-wing aircraft excluding shipboard operations.

7. Electromagnetic interference. MIL-STD-461, requirements CE101, RE101 and RS101 are not applicable.

8. Vibration. Delete 4.6.3.4 of MIL-DTL-32729 in its entirety and replace with the following: "The converter shall be subjected to vibration testing of MIL-STD-810, Method 514, Procedure I (general vibration), Annex D, Category 14 (rotary-wing aircraft - helicopter). The converter shall be operating during the vibration test at full load and 28 VDC input. The output voltage shall be monitored during the test to determine compliance with MIL-DTL-32729, Output Power. The applied vibration reference profile shall be in accordance with MIL-STD-810, Method 514, Annex D, Category 14, helicopter vibration exposure requirements (general materiel) tailored to the AH-1 aircraft platform using the sinusoidal frequencies and magnitudes shown in table III of this detail specification sheet. The duration shall be 4 hours in each of three mutually perpendicular axes. Equivalent methods may be applied to test durations where applicable (i.e., amplitudes may be reduced and durations increased) per guidance provided in MIL-STD-810, Method 514, Annex F, if approved by the Qualifying Activity. The converter shall meet the full performance input voltage requirements of MIL-DTL-32729 during and upon completion of this test sequence and exhibit no structural damage or degradation. Pre- and post-test frequency response measurements shall be used to identify structural damage."

TABLE III. SOR (Sine-on-Random) profile parameters.

Description	Hz	Amplitude
Main Rotor 1P (A_1 /f_1)	4.9	0.12 g
Main Rotor Blade Pass (A_2 /f_2)	19.7	1.97 g
2X Main Rotor Blade Pass (A_3 /f_3)	39.4	2.50 g
3X Main Rotor Blade Pass (A_4 /f_4)	59.196	1.50 g
W_0		0.010 g^2/Hz
W_1		0.010 g^2/Hz
f_t	500	

9. Resonance search. Not applicable.

10. Sinusoidal vibration test. Not applicable.

11. Insulation resistance. The converter shall be tested in accordance with MIL-STD-202-302,

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Test Condition B for a duration of 1 minute.

12. Exported transients. Exported transient testing shall be conducted as part of the Full Performance Transient Input Voltage testing in accordance with MIL-DTL-32729.

13. Lightning induced transient susceptibility. The converter shall be tested in accordance with MIL-STD-461, CS117.

14. Spikes. Spikes testing shall be conducted as part of the Full Performance Transient Input Voltage testing in accordance with MIL-DTL-32729.

15. Humidity. The converter shall only be tested for six 24-hour test cycles.

16. Shock. Functional shock test pulse shall be a sawtooth pulse with a peak pulse of at least 20g's for 11 milliseconds.

17. Water drip. The converter shall be tested in accordance with MIL-STD-810 Method 506 Procedure III. The drip rate shall be 140 L/m^2/hour for 15 minutes with the converter operational.

18. Acceleration. The required acceleration level for all directions shall be 7.0g and shall be applied along three mutually perpendicular axes in two opposite directions along each axis.

NOTES

1. Intended use. The converter specified in this specification sheet is intended for use to convert aircraft 28 VDC power in accordance with MIL-STD-704 to 5 VDC. The converter is intended to be used with USB charging connectors remotely located on the aircraft to provide 5 VDC power for charging peripheral equipment that require 5 VDC USB power for charging. The converter operates strictly as a power supply. Note in requirement 20 (shock) of this specification sheet, crash hazard shock testing is not required. Crash hazard shock testing should be considered in applications where converter breaking loose from mounting during a crash may present a hazard to vehicle occupants and bystanders.

2. USB connector lighting. The remotely located USB connectors may be lighted for ease of use and lighting should conform to MIL-STD-3009 and MIL-DTL-7788 to ensure night vision goggle compatibility. Lighting power should be provided by the 5 VDC connected to the USB connectors.

3. Referenced documents. In addition to MIL-DTL-32729, this detail specification sheet references:

MIL-STD-202-302	Insulation Resistance
MIL-STD-461	Requirements for the Control of Electromagnetic Interference Characteristics of Subsystems and Equipment
MIL-STD-464	Electromagnetic Environmental Effects Requirements for Systems
MIL-STD-704	Aircraft Electric Power Characteristics
MIL-STD-810	Environmental Engineering Considerations and Laboratory Tests
MIL-STD-882	System Safety
MIL-STD-3009	Lighting, Aircraft, Night Vision Imaging System (NVIS) Compatible
MIL-DTL-5541	Chemical Conversion Coatings on Aluminum and Aluminum Alloys
MIL-DTL-7788	Panels, Information, Integrally Illuminated
SAE AS31021	Connector, Receptacle, Electric, Box Mounting, Solder Contacts, AN Type

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CONCLUDING MATERIAL

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Preparing activity:
Navy - AS
(Project 6130-2022-002)

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